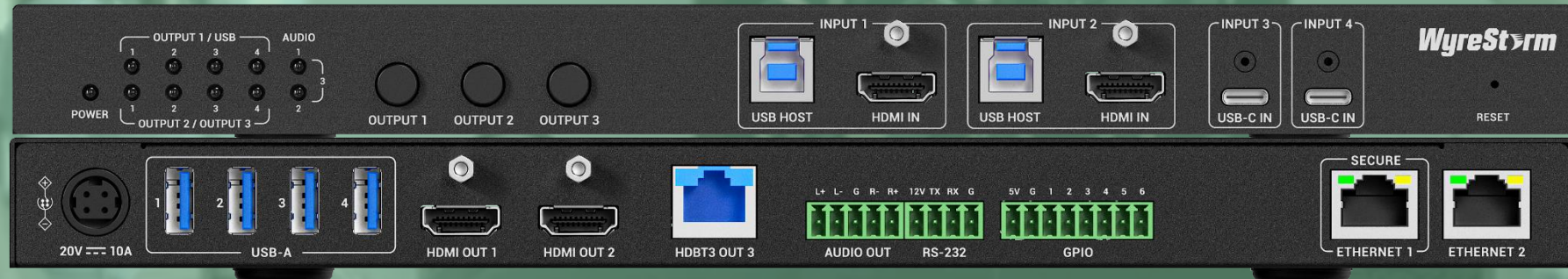


Product Guide

MX-0403-H3-MST

Advanced 4K Conference Presentation Switcher



WyreStorm®

*Award-Winning Audio-Visual
Solutions Manufacturer*

Contents

Product Overview & Main Features

Why Choose Us?

Interface & Specifications

Typical Connection & Applications

Competitor Analysis

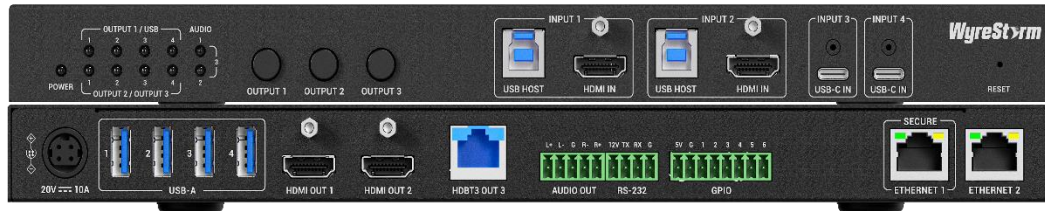
Quick Installation Guide & Web UI Introduction

Q & A Session



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Product Overview

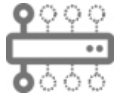


Advanced 4K Conference Presentation Switch

The WyreStorm MX-0403-H3-MST Conference Presentation Switch is the ultimate solution for modern mid-to-large conference rooms, designed to simplify and enhance your presentation experience. This all-in-one device integrates video, audio, long-distance transmission, USB, Ethernet, and control functionalities. With dual USB-C and dual HDMI inputs, automatic source switching, and Multi-Stream Transport (MST) capabilities, the MX-0403-H3-MST delivers an immersive and seamless experience for any conference environment. Its advanced HDBaseT output supports long-distance transmission of up to 4K@60Hz 4:4:4 8-bit over 100m via Cat 6a/7 cable, perfectly suited for medium-to-large spaces.

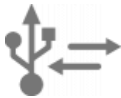
In addition, USB 3.2 ports support transfer speeds of up to 10Gbps, enabling a professional hybrid experience for remote meetings or distance learning. This product is ideal for conference rooms or classrooms seeking an immersive AV environment, advanced hybrid functionality, and versatile control options.

Main Features



Comprehensive Input Options

Offers a variety of inputs, including USB-C and HDMI, ensuring compatibility with devices used in conferences, meetings, and educational settings.



Full-Featured USB-C Support

The USB-C port provides power delivery, network, video, audio and USB transmission through a single cable, simplifying connections and maintaining a clean and tidy workspace.



Long-Distance Transmission (HDBaseT)

Supports long-distance transmission of up to 4K@60Hz 4:4:4 8-bit over 100m via Cat 6a/7 cable. Ensures full compatibility with RX3-100 receivers or other peripheral HDBaseT devices such as projectors and professional displays.



KVM Switch for Display

Includes USB 3.0 (Device and Host) ports for cameras, speakerphones, and keyboards. Perfect for hybrid meetings and education, it simplifies cable management and allows easy switching between devices using a single hub, streamlining workflows.



Centralized Control and Automation

Offers various control methods, such as RS232, GPIO, Telnet, and Web-UI, enabling remote monitoring and control of the presentation system (SYGMA). Provides real-time access to system status, configuration, and performance.



Reliable and High-Quality Resolution

4K resolution (4K@60Hz 4:4:4 8bit) guarantees sharp and clear video, delivering a fully immersive presentation experience.



USB-C MST (Multi-Stream Transport) Supports

Supports USB-C MST technology, enabling simplified connections, reduced cabling, efficient resource utilization, and flexible multi-screen configurations.

Also aligns with the trend toward slimmer, more



Balanced Audio De-embedding

Independent de-embedding audio works seamlessly with external audio systems. When paired with a professional amplifier, speaker, and mic, it creates a more immersive audio experience for presenters and participants.



Versatile and Flexible Ethernet

Advanced Ethernet capabilities provide a reliable network environment, enhancing interactive experiences, especially during hybrid meetings and education. Ethernet Bridge and Ethernet Isolation ensure a stable and secure usage experience.

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Why Choose Us: 1. Immersive Viewing Experience

Deliver an immersive viewing experience with 4K@60Hz 4:4:4 8bit resolution, full-featured USB-C inputs, long-distance transmission, and MST capability, providing high-detail and engaging visuals for the audience.

High-Quality Video Performance

Provide high-quality 4K@60Hz 4:4:4 8bit resolution for detailed information on large-format screens, ensuring clear and reliable AV communication, essential for effective decision-making.

Full-Featured USB-C Inputs

Featuring 2x full-featured USB-C inputs, supporting AV and USB signal transmission, 60W PD charging, and 1G network delivery. Ensures participants enjoy uninterrupted meetings without concerns about power or connectivity.

Long-Distance Transmission

Supports long-distance transmission up to 4K@60Hz 4:4:4 over 100m (Cat 6a/7), simplifying installation and providing an effective solution for medium to large-scale applications.

MST Technology Supports

With USB-C MST (Multi-Stream Transport), connecting multiple displays is effortless, requiring fewer cables and cleaner setups, while aligning with the trend toward slimmer, lighter devices.



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Why Choose Us: 2. Creating Professional Hybrid Environment

Provides effortless connectivity, flexible USB extension, and efficient operation for hybrid conferencing and teaching, enabling all participants to collaborate effectively.

Effortless Connectivity

Works flawlessly with USB3.0 peripherals (cameras, speakerphones, etc.) and seamlessly switches between different host devices (laptops, mini-PCs), delivering a high-quality hybrid experience.

Flexible USB Peripheral Extension

Supports both local and remote USB device connections (HDBT), ensuring greater flexibility for customers in accessing video and audio in hybrid environments.

Efficient Operation in Hybrid

Enhance efficiency of the hybrid meetings with auto-switching features or by integrating with SYN-TOUCH10.

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Why Choose Us: 3. Powerful Ethernet Capability and Security

The advanced network framework supports flexible network settings, versatile network connection, and network security.

Flexible Network Settings

Featuring 2x Ethernet ports, providing Ethernet Bridge, perfect for efficient communication without restrictions; and Ethernet Isolation, ideal for customer to separate internal and guest network connection.

Versatile Network Connection

Seamless network access through multiple interfaces: 2x Ethernet ports, 2x USB Host, 2x USB-C inputs. Reliable network connectivity keeps your meetings smooth and stress-free.

Secure Network Access

Independent Ethernet switching, VLAN settings, and advanced protocols such as HTTPS and 802.1X ensure secure communication and safeguard user privacy.



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Why Choose Us: 4. Effortless Device Management

Centralize and simplify device management to ensure a smooth and efficient operation for all your conferencing needs.

Centralized Control

Manage and monitor all connected devices through an intuitive web interface or all-in-one touchpad controller, simplifying the administration process.

Remote Configuration

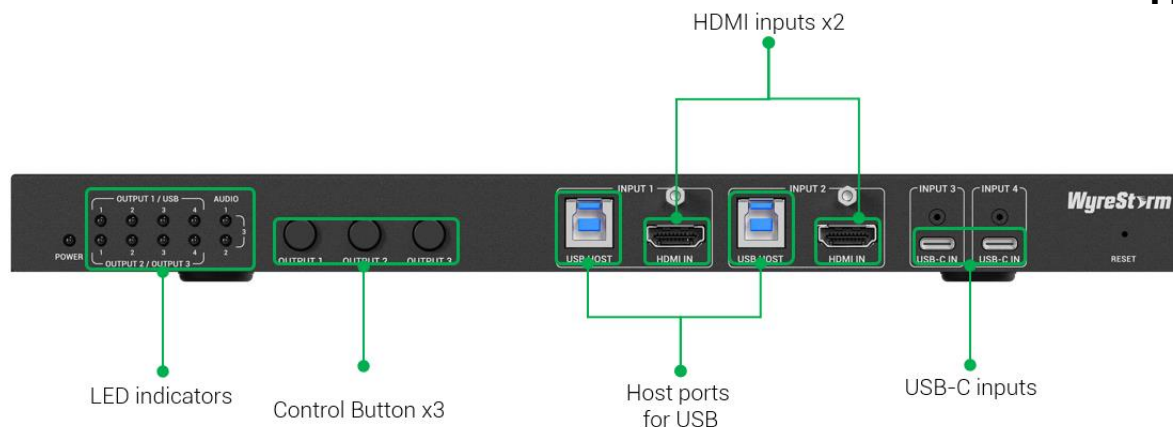
Configure the system remotely via SYGMA cloud device management, ensuring that everything runs smoothly without needing to be physically present.

More Advanced Control

Pair it with SYN-TOUCH10 for intuitive touch control, or with cable cubby (IDB-300) for effortless, seamless operation. Advanced control methods make operation smarter and simpler.

Interface of the Product

Front Panel



Featuring:

2x USB-C Inputs

Ensures secure and stable network access for guests using their PCs, while allowing easy configuration of the system via a web interface.

HDBT Output

Long-distance transmission up to 4K@60Hz4:4:4 over 100m (Cat 6a/7), simplifying installation and providing an effective solution.

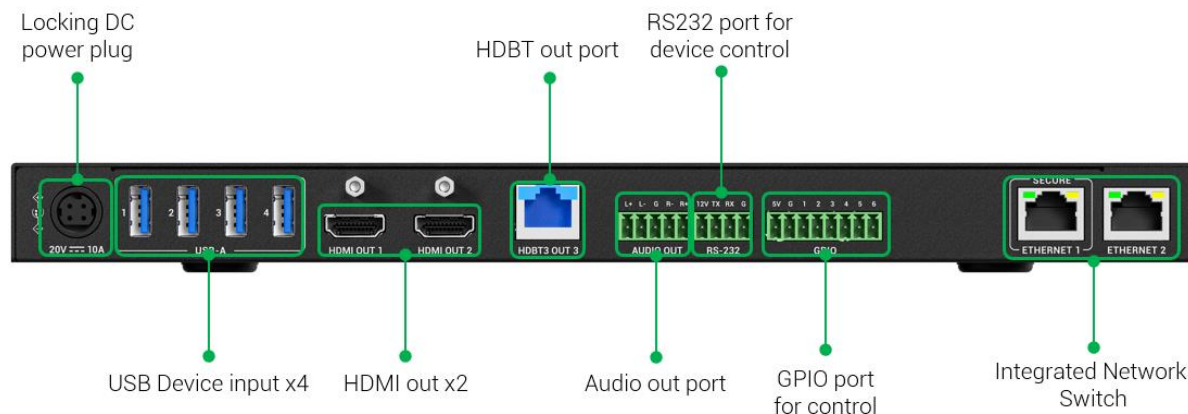
USB Hosts & Devices

Provides direct connection for USB peripherals such as cameras, speakerphones, keyboards, and mice, ensuring seamless local device integration and smooth collaboration.

2x Ethernet Ports

Integrated network switch with 2x Ethernet ports, supporting Ethernet bridge and isolation functions for flexible network deployment and device configuration.

Rear Panel



Specifications & Packing List

Main Specifications	
Input Video Port	2x USB-C, 2x HDMI
Output Video Port	2x HDMI, 1x HDBT
Output Video Signal	HDMI OUT 1: HDMI 2.0, HDCP 2.3 HDMI OUT 2: HDMI 2.0, HDCP 2.3
Input Resolutions (Maximum)	USB-C/HDMI:4K@60Hz 4:4:4 8bit
Output Resolutions (Maximum)	HDMI OUT 1 & 2: 4K@60Hz 4:4:4 8bit HDBT OUT: 4K@60Hz 4:4:4 8bit
Input Audio Port	2x HDMI, 2x USB-C
Output Audio Port	2 x HDMI, 1x HDBT 1 x Analog Audio Out
USB Host	USB Host IN 1&2: USB3.2, up to 10Gbit/s
USB Device	USB3.2, up to 10Gbit/s
Ethernet	1000M/100M adaptive network
Control Method	Front panel buttons, LAN (Web UI & Telnet API), RS-232
Power Supply	20V 10A DC

What's in the Box?

- 1x MX-0403-H3-MST
- 1x 20V/10A Power Supply (US/UK/EU/AU)
- 2x USB-C 2m cables
- 2x USB 3.0 A to B 1.8m Cables
- 4x Mounting Brackets
- 4x Bracket Screws
- 1x 5-pin Phoenix Connector
- 1x 3-pin Phoenix Connector
- 1x 8-pin Phoenix Connector
- 1x Quick Start Guide



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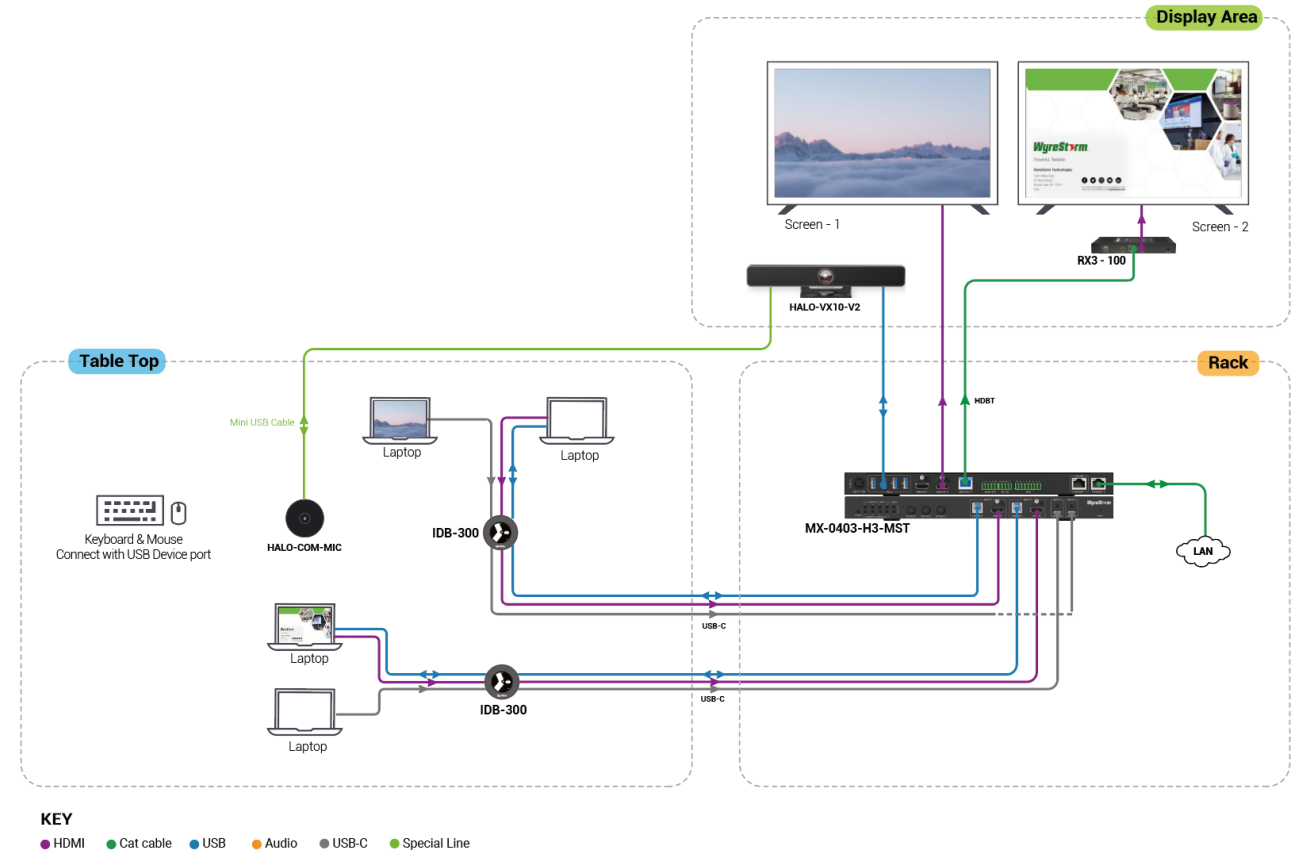
Typical Connection & Applications

Application Scenario 1: Medium Conference Room

Enable seamless collaboration in medium conference rooms by integrating high-quality video, long-distance transmission, hybrid connectivity, and flexible control options. Participants can easily share content, interact with remote colleagues, and enjoy a professional and efficient meeting experience.

Key Benefits Include:

- **High-quality viewing experience:** Deliver up to 4K@60Hz 4:4:4 8bit resolution;
- **Long-distance transmission:** Up to 4K@60Hz 100m (Cat 6a/7), working with RX3-100;
- **Hybrid teaching & meeting :** Connect to videobar, keyboard and mouse, we offer professional hybrid experience.
- **Powerful Ethernet capability:** 2x Ethernet ports for Ethernet bridge and Ethernet isolation, USB Hosts also support Ethernet passthrough;
- **Flexible control methods :** Working with IDB-300 or Web-UI make the control easier ;



Typical Connection & Applications

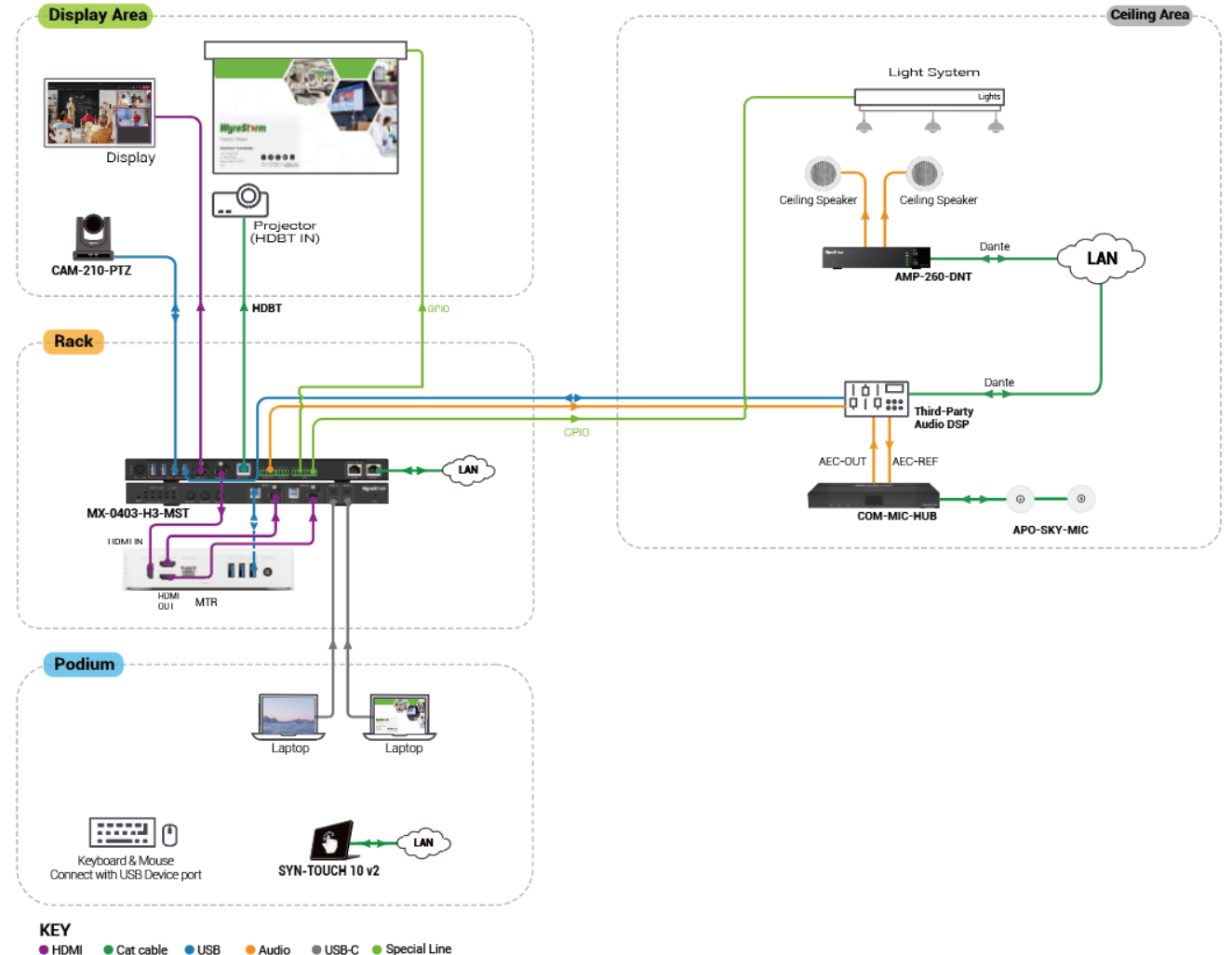
Application Scenario 2:

Medium Conference Room (Hybrid)

Deliver a professional hybrid conferencing solution for medium rooms by combining high-quality video, advanced audio integration, and reliable long-distance transmission. Both in-room and remote participants can clearly see, hear, and interact with each other, creating an efficient and engaging collaboration environment.

Key Benefits Include:

- **High-quality viewing experience:** Deliver up to 4K@60Hz 4:4:4 8bit resolution;
- **Long-distance transmission:** Up to 4K@60Hz 100m (Cat 6a/7), working with RX3-100;
- **Provide professional audio performance:** Integrate with external audio DSPs, amplifiers, microphones, and ceiling speakers to enhance the audio experience
- **Hybrid teaching & meeting :** Connect to CAM-210-PTZ, extra audio system, and keyboard and mouse, we offer professional hybrid experience.
- **Powerful Ethernet capability:** 2x Ethernet ports for Ethernet bridge and Ethernet isolation, USB Hosts also support Ethernet passthrough;
- **Flexible control methods :** Cooperating with Web-UI, SYN- TOUCH10 make the control easier;



History of Previous Products



MX-0402-MST

VIDEO

- **High-quality resolution:** 4K@60Hz 4:4:4 8 bit, HDCP2.3
- **Full-featured USB-C:** supports AV, USB, 60W PD and 1G network to laptop
- **Support MST:** output 2 different images from 1 USB-C laptop to 2 displays
- **Dual Outputs :** (Do not support long-distance transmission [HDBT output] comparing with MX-0403-H3-MST)

USB

- Built-in 4x USB Host (2x USB-C built-in, 2x USB HOST) and 4x USB3.0 ports, ideal for hybrid meeting and hybrid teaching
- USB Host delivers 1G network to PC/laptop

CONTROL

- Work with SYN-TOUCH10 for Smart Control
- RS232, GPIO
- Pair with IDB-300-BTN/IDB-300 for tidy setup and efficient control.

AUDIO

- Balanced De-embedding Audio Out

ETHERNET

- Ethernet Isolation, Ensure Network Security
- Ethernet extension, when there's no enough network cables
- USB HOST supports Ethernet pass through, easy to enjoy the networking

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Comparison with Other Competitors

	WyreStorm		Lightware		
Model	MX-0403-H3-MST	UCX-4x3-HC40-BD (New)	UCX-4x3-TPX-TX20	UCX-4x3-TPN-TX20 (New)	UCX-4x3-HCM40 (New)
HDMI input	2x HDMI, HDMI2.0	2x, HDMI2.0	2x, HDMI2.0	2x, HDMI2.0	2x, HDMI2.0
USB-C	2x USB-C, USB3.2 Gen2 10Gbit/s	2x, USB3.1 Gen1 5Gbit/s	2x, USB3.1 Gen1 5Gbit/s	2x, USB2.0 480Mbit/s	1x, USB3.1 Gen1 5Gbit/s
Output	HDMI Output x 2 HDBT Output x1	HDMI Output x 3	HDMI Output x 2 AVP Output x 1	HDMI Output x 2 AVP Output x 1	HDMI Output * 3
HDBT/AVP	HDBT 3.0 out	No	Yes/AVP * Point to point	Yes/AVP * over Ethernet switch	No
PoE	No	No	Yes	No	No
Dual Screen Expansion (MST)	Yes	No	Yes	No	DP Alt + DisplayLink
Resolution	4K@60Hz 4:4:4 8bit	4K@60Hz 4:4:4 8bit	4K@60Hz 4:4:4 8bit	4K@60Hz 4:4:4 8bit	4K@60Hz 4:4:4 8bit
Audio out	Yes	Yes	Yes	Yes	Yes
No FAN (No noise)	Yes	No	No	No	No
KVM Host	2x USB Host 2x USB-C (USB host built-in)	1x USB Host 1x USB-C	2x USB Host (Type-B)	2x USB Host 2x USB-C	1x USB Host (Type-B)
USB Device	USB 3.0			USB 2.0	
KVM Device	4x USB Device (Type-A)	4x USB Device (Type-A)	4x USB Device (Type-A)	4x USB Device (Type-A)	4x USB Device (Type-A)
Host Switching	Yes	Yes	Yes	Yes	Yes
Ethernet	2x Ethernet	4x: 3x Ethernet, 1x Dante	3x: 3x Ethernet	3x: 3x Ethernet	3x
USB to Ethernet Bridge	All USB-C/B	USB-C only	USB-C only	USB-C only	No
RS232	Yes	Yes	Yes	Yes	Yes
GPIO	Yes	Yes	Yes	Yes	Yes
OCS Sensor (with 24V power supply)	Yes	Yes	No	Yes	Yes
Panel Control Button	Yes	Yes	Yes	Yes	Yes
USB-C Power	60W+60W	100W (120W total)	100W (120W total)	100W (120W total)	100W (120W total)

Quick Installation Guide

Effortlessly set up your presentation switcher with simple steps, ensuring seamless configuration and an exceptional viewing experience.

- **Connect Interfaces:** Follow the standard product connection guidelines and customize connections based on your environment's layout for optimal performance.
- **Start the Device:** Use SmartGUI to locate the device's IP address, then log into the Web-UI for easy equipment configuration.
- **Create Your Hybrid Environment:** Create your video, audio, USB settings using the Web-UI. For detailed instructions, refer to the user manual..
- **Enjoy Superior Visuals:** Experience stunning video clarity and immersive visuals tailored to your needs.



Important Notes for Installation

- Ensure all cables are securely connected to avoid performance issues.
- Check your network settings and confirm that the device is on the correct subnet.
- Double-check configuration settings in the Web-UI to optimize display and functionality.
- For complex installations, refer to the documentation or seek professional assistance if necessary.

Introduction of Web Configuration Interface

The Web UI of the MX-0403-H3-MST presentation switcher is designed for intuitive device management and control through any web browser. With the Web UI, you can easily configure and customize your own settings with the following features:

Web UI Functionality:

- **Interface:** Effortlessly manage device settings via a web browser.
- **Setting up:** Video routing, audio control, and USB settings or other general settings such as Ethernet, RS232, CEC, etc.
- **Control:** Set up and adjust video inputs, outputs, and EDID profiles.

Key Web UI Configurations:

- **General:** Configure video switching, auto-switching, EDID, and HDCP.
- **Control:** Enable RS-232, GPIO for device control and CEC for display control.
- **Network:** Easy setting up about Ethernet capabilities.
- **System:** Configure login password, system log, and perform firmware updates.

Final Thoughts

Q & A Session

Got questions? Don't hesitate to ask away.

